



Computational approach to integrate 3D X-ray microtomography and NMR data

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ABSTRACT

Nowadays, most of the efforts in NMR applied to porous media are dedicated to studying the molecular fluid dynamics within and among the pores. These analyses have a higher complexity due to morphology and chemical composition of rocks, besides dynamic effects as restricted diffusion, diffusional coupling, and exchange processes. Since the translational nuclear spin diffusion in a confined geometry (e.g. pores and fractures) requires specific boundary conditions, the theoretical solutions are restricted to some special problems and, in many cases, computational methods are required. The Random Walk Method is a classic way to simulate self-diffusion along a Digital Porous Medium. Bergman model considers the magnetic relaxation process of the fluid molecules by including a probability rate of magnetization survival under surface interactions. Here we propose a statistical approach to correlate surface magnetic relaxivity with the computational method applied to the NMR relaxation in order to elucidate the relationship between simulated relaxation time and pore size of the Digital Porous Medium. The proposed computational method simulates one- and two-dimensional NMR techniques reproducing, for example, longitudinal and transverse relaxation times (T_1 and T_2 , respectively), diffusion coefficients (D), as well as their correlations. For a good approximation between the numerical and experimental results, it is necessary to preserve the complexity of translational diffusion through the microstructures in the digital rocks. Therefore, we use Digital Porous Media obtained by 3D X-ray microtomography. To validate the method, relaxation times of ideal spherical pores were obtained and compared with the previous determinations by the Brownstein-Tarr model, as well as the computational approach proposed by Bergman. Furthermore, simulated and experimental results of synthetic porous media are compared. These results make evident the potential of computational physics in the analysis of the NMR data for complex porous materials.

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1. Introduction

Nuclear spin relaxation in confined fluids has been extensively used for decades to obtain information about the structure and morphology of porous materials, such as cement, bones, and rocks [1–3], as well as the fluid molecules migration [4–6]. During the relaxation process, the nuclei are able to permeate the medium due to translational diffusion before total magnetization returns to the thermal equilibrium. Therefore, the magnetization decay due to relaxation contains information about the way the nuclei diffused through a porous medium and how they interacted with

its surfaces. All this information is detected through the measurement of longitudinal and transverse relaxation times (T_1 and T_2 , respectively), diffusion coefficients (D), as well as their correlations (T_1 – T_2 , T_2 – T_2 , D – T_2) [7–9].

The multi-dimensional NMR experiments allow investigating the molecular dynamics of the fluid within porous media. However, most analyses are based on the linear relationship between relaxation time and pore size, as defined by Brownstein and Tarr [10] (henceforth referenced as *Brownstein-Tarr* when mentioning their work). As an example, one could mention the determination of the surface magnetic relaxivity, an important parameter in obtaining pore size distributions, from a D – T_2 experiment [11,12]. To get this information from the NMR data it is necessary to apply the linear relationship proposed by *Brownstein-Tarr* to the

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restricted diffusion equations proposed by Mitra et al., reinterpreting this effect from the NMR point of view.

The relaxation time distributions bear the effects of microscopic dynamics. To explore the NMR response in complex materials, different simulations were developed in the past decades. Mendelson et al. [13] and Bergman et al. [14], using a lattice random walk method, simulated the spin relaxation in a saturated porous system, reproducing the dynamics and relaxation process, and will be referenced along the text as *Bergman*. Later, Arns et al. [15,16] and Shikhov et al. [17] applied this model with the aid of 3D X-ray microtomography (micro-CT) images to represent the porous matrix of different samples, exploring the NMR techniques and permeability.

Solving the problem of diffusion as a random walk problem is a well-established problem in the literature, where the method is based in considering that each walker (molecule, nucleus) undergoes successive statistically independent displacements [18]. In this work, the simulation of the nuclear spin magnetization dynamics in a porous medium was modeled based on the spin packet concept, presented by Bergman, Arns, and co-workers. This approach implies that each walker represents a volumetric ensemble of excited nuclear spins. The NMR signal contributions are calculated taking into account the interactions of the nuclei with the pore walls and among themselves.

To preserve the complexity of translational diffusion through the porous medium microstructures, micro-CT images were acquired to represent the Digital Porous Media. Several NMR experiments can be easily reproduced by the developed method, such as CPMG, Inversion-Recovery, Exchange and Diffusion, as well as their correlations [5,7,19,20].

The outcomes can be used to improve the interpretation of the porous media NMR measurements and correlate them with morphological and physicochemical parameters, such as surface magnetic susceptibility, pore size distribution, tortuosity, and permeability.

This work is organized as follows. In the first section, we present a brief theory of NMR in porous media with some examples of the mechanisms that produce relaxation. In the second section, we describe the computational method and how it is used to simulate diffusion and relaxation inside the porous media. In the last section, we present the simulated results and compare them to the experimental data indicating the strengths and shortcomings of the method.

2. NMR theory

Brownstein and Tarr [10] solved the Bloch-Torrey equation [21] for simple confined geometries to study the multiple relaxation times of water in biological cells. They found that, under certain conditions, it is possible to correlate the surface to volume ratio of the structure with the fluid's surface relaxation times, Eq. (1),

$$\frac{1}{T_{1,2}^S} = \rho_{1,2} \frac{S}{V} = \rho_{1,2} \frac{c}{R}, \quad (1)$$

where ρ is the surface relaxivity, S/V is the surface to volume ratio (SVR), defined here as a length scale ($1/R$) of a pore in the porous media, and c is a constant dependent on the pore geometry, here considered as spherical pores ($c = 3$). Eq. (1) is valid in the fast diffusion regime, when the relationship ($\rho R/D_0 \ll 1$) is satisfied; in which D_0 is the bulk diffusion coefficient. In the intermediate and slow regimes, a single pore presents multiple relaxation times and there is no analytical representation of the relaxation times in terms of pore geometry. However, if the surface relaxivity is known, it is possible to numerically solve the transcendental equation that correlates the eigenvalues (hence relaxation times) of the

Brownstein-Tarr equations with the pore length scale (for spherical pores), leading to Eqs. (2) and (3).

$$1 - \Gamma \cot(\Gamma) - \Gamma \left(\rho_{1,2} \sqrt{\frac{T_{1,2}}{D_0}} \right) = 0, \quad (2)$$

$$R = \Gamma \sqrt{T_{1,2} D_0}, \quad (3)$$

in which Γ is the solution of the transcendental Eq. (2), used to determine R in Eq. (3), when the relaxation time and bulk diffusion are known (from an experiment or simulation for example). In general, considering the bulk contribution, the total relaxation rates are defined by

$$\frac{1}{T_1} = \frac{1}{T_1^{\text{Surface}}} + \frac{1}{T_1^{\text{Bulk}}} = \rho_1 \frac{S}{V} + \frac{1}{T_1^{\text{Bulk}}}, \quad (4)$$

$$\frac{1}{T_2} = \frac{1}{T_2^{\text{Surface}}} + \frac{1}{T_2^{\text{Bulk}}} = \rho_2 \frac{S}{V} + \frac{1}{T_2^{\text{Bulk}}}, \quad (5)$$

considering that fast diffusion approximation was assumed for the right-hand side of Eqs. (4) and (5).

2.1. Longitudinal and transverse surface relaxivities: ρ_1 and ρ_2

In this section, we will discuss surface and bulk contributions to relaxation, disregarding effects on transverse relaxation generated by magnetic susceptibility present in most materials, especially in the case of rocks.

According to the Bloembergen-Purcell-Pound (BPP) model [22], a good approximation for magnetic relaxation of a fluid containing spin 1/2 nuclei associates longitudinal relaxation to the repopulation of quantum states, and transverse relaxation to both repopulation of states and loss of coherence. Therefore, for a substance in which thermal motion provides a minimum loss of coherence (as in a low viscosity fluid), the transverse relaxation will be primarily due to the repopulation of quantum states, leading to $T_2 = T_1$. An example of this is bulk water. According to this model, molecules close to the surface may experience dynamic restrictions due to affinity issues, mainly in the translational and rotational movements, which induce different surface relaxation rates leading in most cases to $T_1^{\text{Surface}} \neq T_2^{\text{Surface}}$.

Additionally, Kleinberg et al. [23] described the mechanism for the $T_{1,2}^{\text{Surface}}$ by the molecular motion close to the surface, in a layer assumed to be about the size of a single water molecule. They defined that, in the case pore spaces of sedimentary rocks at low magnetic field, the relaxation mechanisms associated with pure diamagnetic sites on the surface and by nuclear-nuclear dipolar coupling modulated by hindered molecular reorientation do not dominate the NMR relaxation, and the paramagnetic relaxation mechanism is the main contribution for the relaxation on surface.

Korb and co-workers [24] proposed that motion of water molecules within the layer near the surface consists of a two-dimensional random walk and the surface is composed of different sites (magnetic and non-magnetic). The exchange processes between these sites and between the layers close to the surface and the bulk phase of the fluid are characterized by the times τ_m and τ_s , respectively. This theoretical approach allows analyzing the dependence of the surface relaxation versus frequency by the field-cycling magnetic relaxation measurements [24].

Independent of the relaxation mechanism on the surface, the proportionality between surface relaxation time and SVR is found. Song [2] and McDonald et al. [1], presented the relaxation on surface as a layer δ of the fluid, with volume δ , S , that relaxes by the surface mechanism and with a characteristic time $T_{1,2}^{\text{Surface}}$. Considering the total surface S of the pore, and taking the average of the

total molecules that diffuse between this thin layer and the bulk phase of the pores (total pore volume of V_p), the relaxation time is:

$$\frac{1}{T_{1,2}} = \frac{\delta S}{V_p} \frac{1}{T_{1,2}^{\text{Surface}}} + \frac{1}{T_{1,2}^{\text{Bulk}}}, \quad (6)$$

where considering Eq. (1), the relaxivity is given by $\rho_{1,2} = \delta/T_{1,2}^{\text{Surface}}$. The dependence of the $T_{1,2}^{\text{Surface}}$ with the density functions for the motion $J_L^{0,1,2}(\omega)$, detailed elsewhere [24], gives the information about the effects of repopulation and decoherence mechanisms on the surface in the molecular scale, where δ is assumed to be about the size of a single water molecule.

However, here, we will consider that surface relaxation is due to the average contribution of all the molecules that diffuse between the thin layer near the pore surfaces (nanoscale) and the bulk phase, Eq. (6). This approximation is necessary in order to simulate the magnetization evolution in a porous material with the aid of micro-CT images, where the size of the Digital Porous Medium is a few millimeters, which preserves the pore morphology.

The proposal is to write the mechanism of repopulation and decoherence by different terms in the surface magnetic relaxivity. It is known that the transverse relaxivity (ρ_2) contains both mechanisms (repopulation and decoherence) that drive the transverse magnetization to zero. However, the longitudinal contribution ρ_1 , on the other hand, is exclusively dependent on repopulation of states on the surface. Therefore, we propose to rewrite the relaxivity ρ_2 as the sum of the two mechanisms:

$$\frac{1}{T_2} = (\rho_1 + \rho_{\text{phase}}) \frac{S}{V} + \frac{1}{T_1^{\text{Bulk}}}, \quad (7)$$

in which ρ_{phase} is the surface relaxivity exclusively due to the loss of coherence.

Fig. 1a shows the T_1 - T_2 correlation. Estimation of larger pore sizes is limited by T_{Bulk} , as in this regime the relaxation contribution due to the surface interaction is negligible compared with the relaxation among the fluid molecules (bulk). Fig. 1b indicates the dependence of the pore size and the free path of molecules during the respective transverse relaxation time, $\lambda_{T_2} = \sqrt{6DT_2}$. When, $(R/\lambda_{T_2}) \geq 1$, the relaxation time can be considered as originating exclusively from the bulk relaxation and T_2 has no information about the pore size. The other limits are the case $(R/\lambda_{T_2}) \leq 0.1$, in which the relaxation is due to the surface relaxivity, and the pore size can be estimated, and $0.1 \leq (R/\lambda_{T_2}) \leq 1$, in which the pore size can be estimated by subtracting the bulk contribution. However, estimation in the latter case is not well defined. Fig. 1c shows the regimes described by Brownstein-Tarr: fast, intermediate and slow diffusion, defined for three different limits of the relation $(\rho_{1,2}R/D_0)$. In the limit of $(\rho_{1,2}R/D_0) \gg 1$, the system is considered in a slow diffusion regime, and it indicates the eigen-modes should appear. In this same regime the bulk relaxation is predominant; therefore, measuring pore sizes in the slow diffusion regime is compromised for both effects.

Eq. (7) indicates that only for $\rho_{\text{phase}} \gg \rho_1$ the contribution due to the loss of energy (quantum states repopulation) is negligible compared to the decoherence. Therefore, considering the ρ_2 estimated in the NMR experiments and SVR by Eqs. (1) and (3), one way to individually identify the ρ_1 and ρ_{phase} is measuring the T_1 to the T_2 ratio to obtain first ρ_1 . Considering water saturated porous media, $T_1^{\text{Bulk}} = T_2^{\text{Bulk}}$, the ratio can be rewritten as:

$$\frac{T_1}{T_2} = \frac{T_1^{\text{Bulk}} \rho_2 (S/V) + 1}{T_1^{\text{Bulk}} \rho_1 (S/V) + 1}, \quad (8)$$

rewriting to obtain ρ_1 :

$$\rho_1 = \rho_2 \frac{T_2}{T_1} + \frac{1}{T_1^{\text{Bulk}}} \frac{V}{S} \left(\frac{T_2}{T_1} - 1 \right), \quad (9)$$

and which this information to obtain ρ_{phase} . The surface relaxivity and SVR can be estimated by NMR experiments (D- T_2 experiment) or by complementary methods, such as, mercury injection capillary pressure (MICP) and microtomography images correlated with relaxation NMR data. Eq. (9) shows the dependence of the surface relaxivity with the bulk relaxation and pore size, where for smaller pores and longer bulk relaxation the contribution is directly proportional to the surface mechanism of relaxation. It is also clear that when $T_1 = T_2$ in porous media, it implies that $\rho_1 = \rho_2$, meaning that the main contribution on the surface is related to the spectral function with the following frequency dependence: $J_L^{1,2}(\omega)$.

3. Methods

Based on the random walk method (RW) to simulate the translational diffusion of the nuclei in digital rock samples, we propose a simple formalism to integrate the computational parameters that involve NMR magnetization decay and the surface relaxivity. The sample is described by a 3D digital image obtained by micro-CT, and segmented in pores and solid materials.

In this work, the magnetization evolution will be simulated inside a Digital Porous Media obtained from synthetic samples: Ceramic and Glass Bead. The Digital Porous Media was obtained using a Bruker SkyScan 1272 micro-CT, with filters of Al (80 kV, 140 μ A) and Al-Cu (90 kV, 120 μ A), respectively.

These samples have a clear threshold that identifies the solid matrix and void space (pores), therefore, in this case, simple filters and the Interactive Threshold Method were applied to segment the Digital Porous Media in Solid and void space (porous) [25]. These methods were applied using the software provided by SkyScan/Bruker (CTan).

However, in sandstones and, especially in carbonates, this threshold is not easy to define, and it is necessary to use more sophisticated techniques to process the image and identify the pores correctly. An example of a filter method is the noise reduction algorithm Non-Local Means, which does not interfere significantly in the surface of the pores when applied [26]. Other possible solution, is the identification of the interface between the pore and solid matrix, for example, by a gradient of the image, where the interface can be locked during the segmentation process and combined with others advanced methods to provide a better Digital Porous Medium.

3.1. NMR relaxation

The magnetization decay will be characterized by the two relaxation process: bulk and surface relaxation. To connect the computational model with the bulk and surface relaxation times, we will first introduce the bulk component and later, with similar concepts, demonstrate the modeling of the surface.

3.1.1. T_1 Relaxation: bulk

An NMR relaxation experiment consists in observing the decaying behavior of the magnetization as it returns to its equilibrium state after an excitation produced by a radio frequency pulse sequence. In order to describe the relaxation of a system of N nuclei under a static uniform magnetic field B_0 , the relaxation will be expressed in terms of the probability rate p_{ij} of a nucleus decaying from energy state i to the state j . The hydrogen nucleus, ^1H (spin 1/2), has two possible energy states, +1/2 and -1/2, and the sum

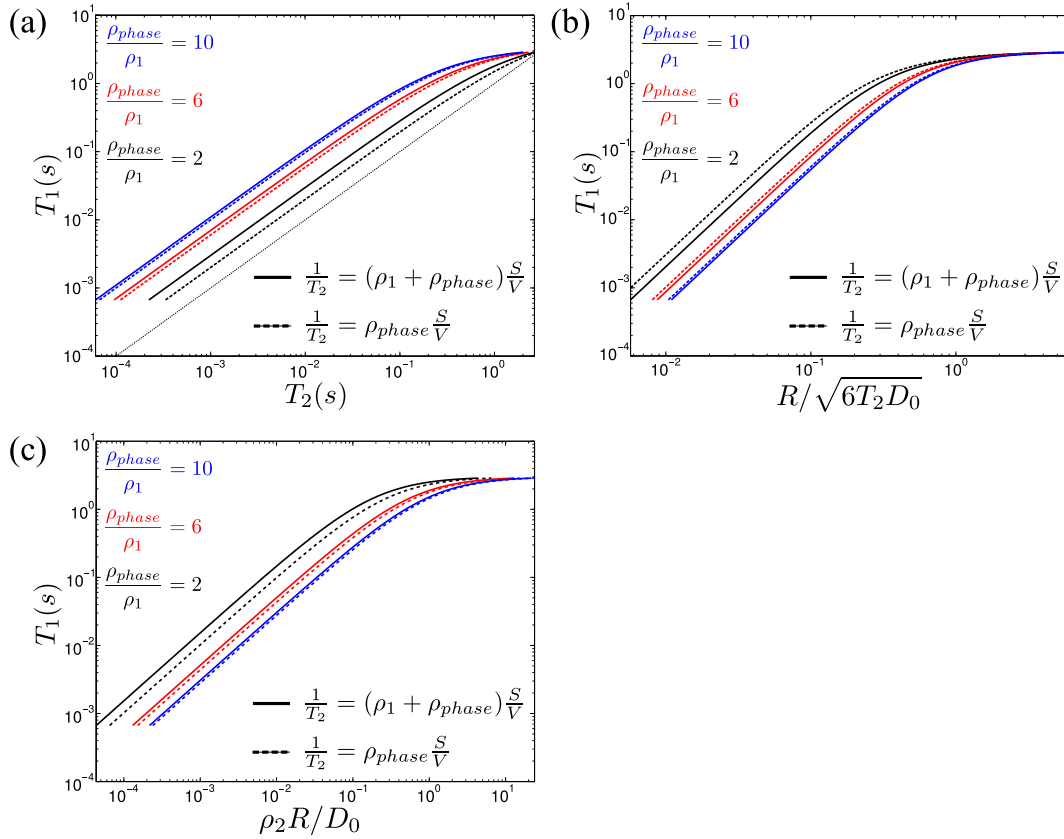


Fig. 1. (a) T_1 - T_2 relaxation maps for three different rates of longitudinal surface magnetic relaxivity (ρ_1) and transverse surface magnetic relaxivity (ρ_{phase}). Note that for $\rho_{phase} \geq 10\rho_1$ the transverse relaxation is predominant in the transverse plane and ρ_2 is exclusively due to loss of phase coherence. (b) The transverse relaxation time is converted to a ratio of the pore radius and the mean square displacement of the particle during the transverse relaxation time. (c) The x-axis is the constant proposed the Brownstein-Tarr to define the diffusion regime. The fast diffusion regime is observed when pore sizes are smaller than the mean square displacement and is mainly due to the surface and not the bulk relaxation.

of all nuclei contribution results in a macroscopic magnetization $M(t)$ [27].

$$M(0) = \sum_i m_i, \quad (10)$$

in which $m = 1$ (low energy state) or $m = -1$ (high energy state). Each of the excited nuclei will then have a probability per unit time p_i of returning to the ground state, and considering the nuclei are indistinguishable, the rate of transition p_i is equal for all of them, yielding the magnetization time evolution:

$$\frac{dM}{dt} = \sum_i -p_i m_i = -p \sum_i m_i = -pM_0. \quad (11)$$

Hence,

$$M(t) = M(0)e^{-p} = M(0)e^{-t/T_1}, \quad (12)$$

where the last part of the equation is the Bloch solution [28]. As a result, the *bulk* longitudinal relaxation is simulated by a random process, with time step Δt , and probability $p = \Delta t/T_1$.

3.1.2. T_2 Relaxation: bulk

In addition to the longitudinal relaxation, the random fluctuations of the magnetic field make the Larmor frequency of each spin fluctuate around its original value, resulting in a decoherence process for an ensemble of spins initially with the same phase. A decoherence process can be simulated by associating to each random walker a phase factor $e^{-i\varphi}$ that can randomly vary for each time step Δt .

From the central limit theorem, if random phases φ_i are accumulated, over all the nuclei after n steps, the total phase $\Phi_i = \sum_i^n \varphi_i$ is normally distributed around its average value [18]. Since the average value is zero, the probability of finding spins with phase Φ is a centered Gaussian [29]:

$$P(\Phi) = \frac{e^{-\Phi^2/(2\langle\Phi_i^2\rangle)}}{\sqrt{2\pi\langle\Phi_i^2\rangle}}. \quad (13)$$

Here, $\Phi^2 = n\langle\varphi^2\rangle$ is the variance of the distribution. The macroscopic magnetization is the average over all nuclear ensemble contribution, and after n steps we expect:

$$M_0 e^{-\frac{\Phi^2}{2}} = M_0 e^{-\frac{n\langle\varphi^2\rangle}{2}}. \quad (14)$$

Considering a time evolution t during n equal steps Δt , and the transverse magnetization given by the Bloch equation, the relation between T_2 and the variance of the probability distribution phase can be written as:

$$M = M_0 e^{-\frac{t}{T_2}} = M_0 e^{-\frac{n\Delta t}{T_2}} = M_0 e^{-\frac{n\langle\varphi^2\rangle}{2}}, \quad (15)$$

$$\langle\varphi_i^2\rangle = \frac{2\Delta t}{T_2}, \quad (16)$$

which defines the probability distribution from which each nucleus loses coherence among themselves.

3.1.3. Surface relaxation

To simulate the relaxation of an ensemble of nuclei in the bulk fluid, the parameters p and $\langle \phi_i^2 \rangle$ can be calculated, for example, using the BPP model [22], or measured experimentally by the user. However, to simulate the evolution of magnetization in complex porous media, it is fundamental to preserve the molecular dynamics, the structure of the sample and the magnetic fluid/surface interaction [24]. Here, the loss of magnetization by the surface interaction is defined by a relaxivity $\rho_{1,2}$, as defined by the Brownstein-Tarr formalism [10].

In each step of the simulation, the nuclei receive a random new position in agreement with the bulk diffusion, and the bulk relaxation is applied. However, if the new position is identified as a collision with the surface the nuclei receive a longitudinal and transverse surface penalization.

The method developed by Bergman uses a lattice random walk method to simulate the spin relaxation of a saturated pore. For this purpose, they define a probability v of a walker dying or surviving an attempt to move towards a solid voxel of the porous media. The probability used in this case is related to the surface relaxivity ρ via,

$$Av = \frac{\rho \varepsilon}{D} + O\left(\left(\frac{\rho \varepsilon}{D}\right)^2\right), \quad (17)$$

which A is a correction factor of order 1 accounting for the details of the random walk implementation, and calculated by Bergman as $A = 3/2$. ε is the random walk parameter, related to the time discretization and diffusion coefficient D_0 , given by:

$$D_0 = \frac{\varepsilon^2}{6\Delta t}. \quad (18)$$

Eq. (18) is determined considering the 3D unrestricted diffusion. Computationally, if the nuclei are in a confined pore, the new position after diffusing during the time step Δt has a chance of lying outside the pore space and within the solid matrix. Two methods to return the nuclei to the pore space, in this case, are: (i) to return it to the last position (retro-reflection) or (ii) to select a new random position (diffusive-reflection). **In situations in which there are diffusion and flow, a third method needs to be applied, the specular-reflection.** As fluid flow analysis is out of the scope of this work, retro-reflection was chosen for dealing with this contour condition.

Based on the theory used in the bulk case, the nucleus that interacts with the surface in the interval Δt , has a rate q_s related to the longitudinal relaxation and a variance $\langle \alpha_{i,s}^2 \rangle$ for the Gaussian probability to gain phase.

We propose to find these parameters as a function of the surface magnetic relaxivity ρ . For this, considering the time evolution t during n equal steps Δt , a set of nuclei accumulate a different phase α due to M collisions, and loses energy with a rate q . The variance α^2 and q in for wall collisions is defined by:

$$\langle \alpha^2 \rangle = \frac{2}{T_2} \frac{t}{M} = \frac{2}{T_2} \tau, \quad (19)$$

$$q = \frac{1}{T_1} \frac{t}{M} = \frac{\tau}{T_1}, \quad (20)$$

the average time between collisions is defined as $\tau = t/M$. As established by Brownstein-Tarr, the relation between T_1 , T_2 and ρ is only analytically established in the fast diffusion limit, when Eqs. (19) and (20) can be rewritten as:

$$\langle \alpha^2 \rangle = 2\rho_{\text{phase}} \frac{S}{V} \tau, \quad (21)$$

$$q = \rho_1 \frac{S}{V} \tau. \quad (22)$$

For complex porous media, the SVR and average collision time are not simply defined. In order to overcome this issue, we propose the following approximation for the relation between the collision time τ and SVR.

For a time $t = n\Delta t$, the total number of nuclei that collide with the surface in each step Δt is approximately:

$$\frac{M}{n} = \tilde{\lambda} \frac{S}{V}, \quad (23)$$

where $\tilde{\lambda}$ is a small layer close to the surface, assumed to be about the size of the diffusion length during a time longer than the characteristic time of the molecular dynamics (see Fig. 2).

Using the diffusion length commonly defined by the Standard Deviation and considering only the nuclei that move toward the surface, we take $\tilde{\lambda} = \sqrt{2D\Delta t}/2$.

However, calculating the mean displacement of the nuclei that move towards the surface, given by:

$$\tilde{\lambda} = \langle x \rangle|_{0 \rightarrow +\infty} = \frac{1}{\sigma\sqrt{2\pi}} \int_0^{+\infty} x e^{-\frac{x^2}{2\sigma^2}} dx = \sqrt{\frac{\sigma^2}{2\pi}}, \quad (24)$$

the distance is $\tilde{\lambda} = \sqrt{D\Delta t/\pi}$, in which the walker moves in the normal direction (1D movement) and, therefore, $\sigma^2 = 2D\Delta t$, considering the 1D unrestricted diffusion.

Hence, the variance and probability of energy loss at the surface can be related to surface relaxivity using the standard deviation by ($\sqrt{\langle r^2 \rangle}$ model):

$$\langle \alpha_i^2 \rangle = 2\rho_{\text{phase}} \frac{S}{V} \left(\frac{t}{M} \right) = \frac{4\rho_2 \Delta t}{\sqrt{2D\Delta t}}, \quad (25)$$

$$q = \rho_1 \frac{S}{V} \left(\frac{t}{M} \right) = \frac{2\rho_1 \Delta t}{\sqrt{2D\Delta t}}, \quad (26)$$

or by the mean displacement ($\langle r_n \rangle$ model):

$$\langle \alpha_i^2 \rangle = 2\rho_{\text{phase}} \frac{S}{V} \left(\frac{t}{M} \right) = \frac{2\rho_2 \Delta t}{\sqrt{D\Delta t/\pi}}, \quad (27)$$

$$q = \rho_1 \frac{S}{V} \left(\frac{t}{M} \right) = \frac{\rho_1 \Delta t}{\sqrt{D\Delta t/\pi}}. \quad (28)$$

These two approximations are compared in Fig. 3 with the standard Bergman model. To compare, we simulated the magnetization decay for a set of spherical pores, with the defined radius R ranging from 5 to 300 μm , and applied the Inverse Laplace Transform in order to obtain the relaxation time distribution. By the general Eqs. (2) and (3), we calculated the experimental radius (R_{exp}) for the known relaxation time and surface magnetic relaxivity ($\rho_2 = 40 \mu\text{m/s}$).

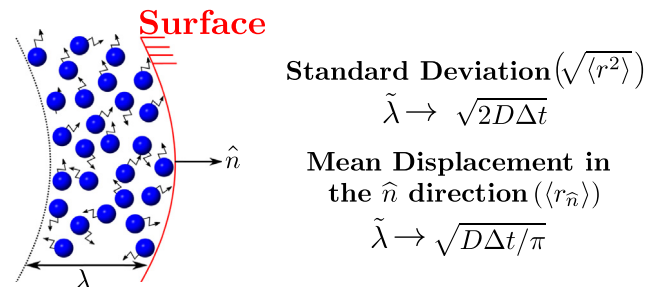


Fig. 2. In a short time interval Δt , the nuclei move in the normal direction of the surface, where, in the layer near the surface S , with volume $V = \tilde{\lambda}S$, half of the nuclei interact. We defined the mean displacement by two different methods, named as ($\sqrt{\langle r^2 \rangle}$) and ($\langle r_n \rangle$).

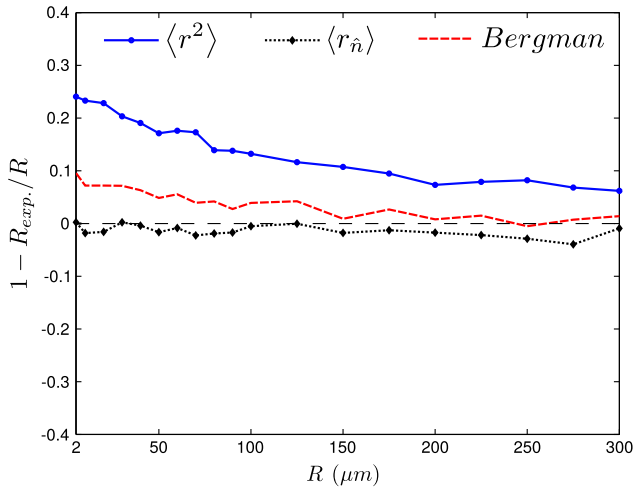


Fig. 3. The radius prediction by the NMR relaxation model, using a standard Bergman and the models proposed here. Each model uses a different method to calculate the average distance to the wall with which the nuclei have a chance to interact. The standard deviation model provides $\bar{\lambda} = \sqrt{2D\Delta t}/2$, and the mean displacement model gives $\bar{\lambda} = \sqrt{D\Delta t/\pi}$, taking into account the movement in the direction normal to the surface S .

Fig. 3 shows the difference between the experimental and the defined radius ($R_{exp} - R$). Independent of the size of the pore and diffusion regime, the model $\langle r_{\hat{n}} \rangle$ presents an expected radius with a better agreement with the defined radius.

4. Results and discussions

To compare the results of our proposed method with that expected by the Bergman model, we present here a comparison between the analytical results from the Brownstein-Tarr model and our simulated results.

Furthermore, we constructed two synthetic samples to perform the NMR experiments and simulations. The first one consists of a ceramic sample of alumina, where pores are produced by saccharose crystals which burn out at high temperature [30]. The grains of saccharose are selected by different sieves to control the porosity and size distribution. The second sample is built packing glass beads and sintering at a controlled temperature until necks are formed. This process provides a rigid glass beads sample that ensures it will exhibit the same morphology for both the NMR experiments and NMR simulations. Both cylindrical samples have 10 mm in diameter and 20 mm length.

For the NMR experiments, it was used a Tecmag LapNMR spectrometer and a 0.47 T permanent magnet, at 30 °C, with a Larmor frequency of 20 MHz for ^1H nuclei. The samples were saturated with distilled water, using a vacuum pressure of 90 kPa, where the samples remained in the water for 60 min.

4.1. Brownstein-Tarr and simulation

To analyze the correspondence between the computational method proposed and the formalism of Brownstein-Tarr, we simulated the magnetization evolution in a set of spheres to verify the presence of fast and slow regimes.

Fig. 4 illustrates the results of the CPMG simulations in the spherical pores, setting the relaxivity as $\rho_2 = 40 \mu\text{m/s}$, bulk diffusion coefficient $D_0 = 2.3 \times 10^{-3} \mu\text{m}^2/\text{s}$ and radii ranging from 2 to 300 μm . This figure shows the distributions of the transverse relaxation times obtained by the Laplace Inversion Method. In this range of radii we see results in the fast regime (blue) and intermediary/slow regime (red), which is easily identified because of the presence of multiple relaxation times for a single pore.

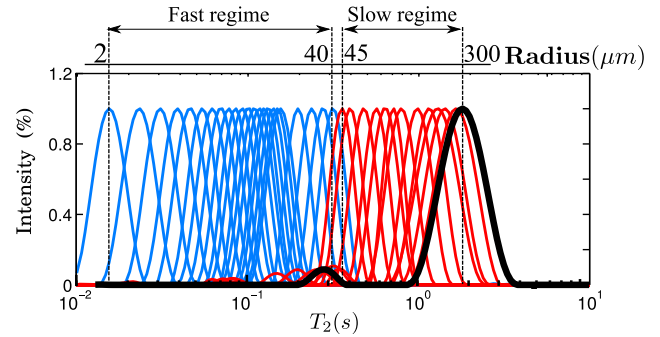


Fig. 4. The CPMG Decays are analyzed by Inverse Laplace transform, resulting in transverse relaxation time distributions that minimize the variance of the fitting. We simulated the CPMG for spherical pores in a range of radii ranging from 2 to 300 μm . The simulated curves for spheres with radii smaller than 40 μm present a single peak, which indicate that the dynamics are in the fast diffusion regime (blue lines). Simulated curves for radii larger than 40 μm have two or more peaks, indicating that, in these cases, the dynamics occur under the intermediate diffusion regime (red lines). The highlighted curve in black, for 300- μm pore, clearly shows the first order normal mode. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

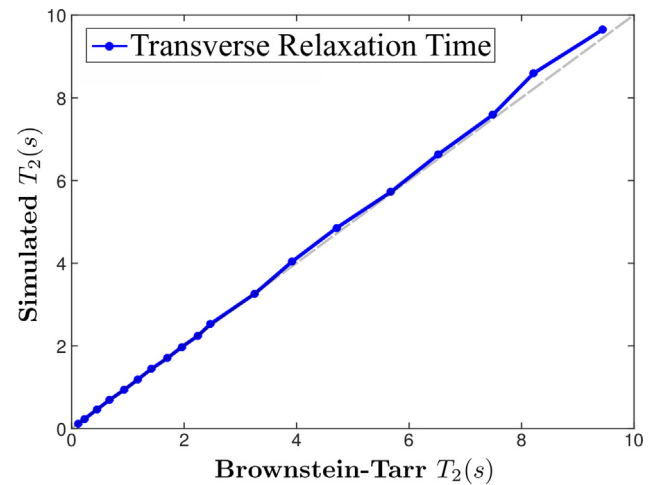


Fig. 5. Comparison between the relaxation time predicted by Brownstein-Tarr and our simulation. With a linear regression, we obtained an $R^2 = 0.99$. For the cases of long relaxation times, the system is under the slow or intermediate regimes, in which the multi-exponential decays due to the eigen-modes occur. Consequently, fittings present greater deviation between the simulated and theoretical relaxation times.

To verify the correlation between the theory and the model we compare the main peak position in the relaxation time distribution with the relaxation time expected from the Brownstein-Tarr solution for the eigenvalue of order zero (see Fig. 5).

Using Eq. (1), we can calculate the surface magnetic relaxivity under the fast diffusion approximation. In Fig. 6, we observe that the relaxivity ρ_2 remains constant for small radii, but decreases as the radius increases because the fast diffusion approximation is not appropriate for the intermediary/slow regime the system starts to assume, in agreement with the theory. The multiple peaks in the relaxation time distribution appear as we get to pores larger than 40 μm , and, at this point, the relaxivity ρ_2 is 7% lower than the value chosen for the simulation in this case.

4.2. Ceramic sample

To test the model in a Digital Porous Medium, a ceramic sample was built with three different pore sizes, ranging from 0.1 to 100

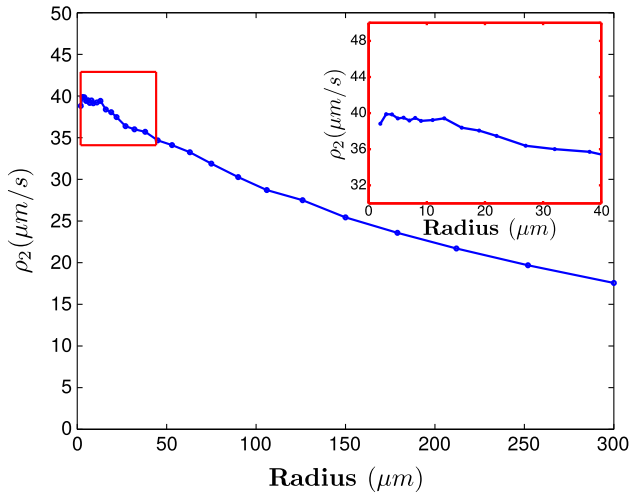


Fig. 6. The relaxivity ρ_2 estimated for different radii diverge for spheres larger than 40 μm because the diffusion regime of the system changes from fast to intermediate. That is verified with $\rho_2 R/D = 0.7$ for a sphere with $R = 40 \mu\text{m}$, $\rho_2 = 40 \mu\text{m/s}$, and $D_0 = 2.3 \times 10^3 \mu\text{m}^2/\text{s}$.

μm orders of magnitude [30]. However, the images used to construct the digital porous medium were performed with 2.5 μm of resolution. Due to this resolution limitation, just two pore sizes appear in the simulation. The matrix block was created with $(1000 \times 1000 \times 1500) \text{ pixel}^3$, which represent 23.45 mm^3 or 5% of the physical sample, Fig. 7.

One of the most important parameters estimated from these simulations is the surface relaxivity $\rho_{1,2}$. To estimate the surface relaxivity, we used as reference the longer experimental relaxation times because the micro-CT image precision for the larger pores is higher. The experimental relaxation time distribution was measured by CPMG, with 24,000 echoes, echo time of 250 μs and a total decay measurement of 6 s. The T_1 - T_2 experiment was performed by the Inversion-Recovery/CPMG, 30 points were chosen from 500 μs to 20 s for the T_1 dimension, and 24,000 echoes with echo time of 250 μs for the CPMG.

d'Eurydice et al. [18] reported a complete characterization of a similar sample and performed the experiments in a 2 T Oxford superconductor magnet. This was possible because of the weak magnetic susceptibility presented by this synthetic sample. We performed the NMR experiments in a 0.5 T magnet, where the effects of the internal magnetic field gradient are negligible.

Starting from an arbitrary relaxivity $\rho_{1,2}$, it is possible to estimate the pore size distribution of the sample. Then, using the experimental relaxation time information from the peaks with longer decay times, we can calculate a first approximation of the

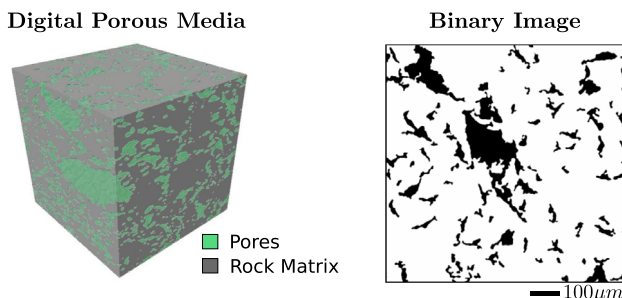


Fig. 7. Ceramic sample with controlled pore sizes to analyze via NMR Experiments and Numerical Simulations. The micro-CT images were performed with a resolution of 2.5 μm and the intrinsic pores, with dimensions below the micrometer scale, are not visible. For the simulations, the 3D digital porous media is constructed.

surface relaxivity with Eq. (1). In the simulated relaxation times distribution, there are two peaks, the longer one corresponding to pores with radii of the order of 60 μm and the shorter one corresponding to pores with 10 μm . The experimental results contain the same peaks, but including a shorter one. This shorter peak corresponds to pores of size on the order of micrometers, which is close to or smaller than the resolution used in the micro-CT images, 2.5 μm . For this reason, the shortest peak is absent in the simulations, Fig. 8(a). For the ceramics, assuming the pores are sphere-like, we found $\rho_2 = 17 \mu\text{m/s}$. Fig. 8(b) shows the pore size distribution calculated using the fast diffusion equation.

Fig. 9 shows the simulated and experimental T_1 - T_2 correlation maps. For the large pores in the ceramic sample, using $T_1/T_2 = 1.8$, Eqs. (8) and (9) provide $\rho_1 = 6 \mu\text{m/s}$ and $\rho_{\text{phase}} = 11 \mu\text{m/s}$. Here, assuming the spherical pores approximation and using a moderate surface relaxivity it was found, for the bigger pore, a radius about 60 μm , Fig. 8. The bulk water relaxation time used was equal to 3 s. The T_1 - T_2 correlation reveals that T_1 to T_2 ratio is not a single constant value and is bigger at shorter relaxation times. This is a consequence of bulk and surface relaxations being of comparable magnitudes for the larger pores, and surface relaxation dominating on smaller pore scales, Eqs. (4) and (5).

4.3. Glass beads sample and slow/intermediate diffusion

The ceramic sample presented in the last section has three well defined pore distributions. The smaller pores have sizes below the micro-CT resolution. Due to this reason, the results from the simulation and experimental data agree only for relaxation times corresponding to pores that can be measured with the 2.5- μm micro-CT resolution. Selecting glass beads ranging from 250 to 300 μm , we produced a second synthetic porous media with larger pores, which are well-defined by micro-CT.

We sintered the glass beads by controlling the temperature until it formed the necks. The glass beads and pores are larger than the micro-CT resolution (2.5 μm), which accounts for a better agreement between the simulated and experimental results. The

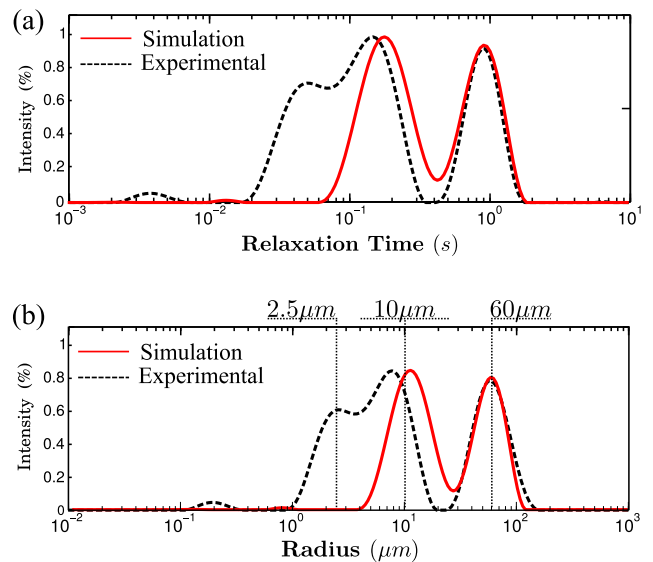


Fig. 8. The NMR experiment was performed in a 0.5 T magnet with a Tecmag spectrometer, and the analysis of the CPMG Decay was made using the Laplace Inversion Method. (a) The T_2 distribution for the simulations of the 10^6 nuclei and the experimental results are in good agreement. The absence of the shortest peak in the simulated data compared to the experimental data can be explained by the size of the pores which produce that peak. Their size is of the order of the micro-CT resolution, and the radii distribution in the figure (b) evidences that.

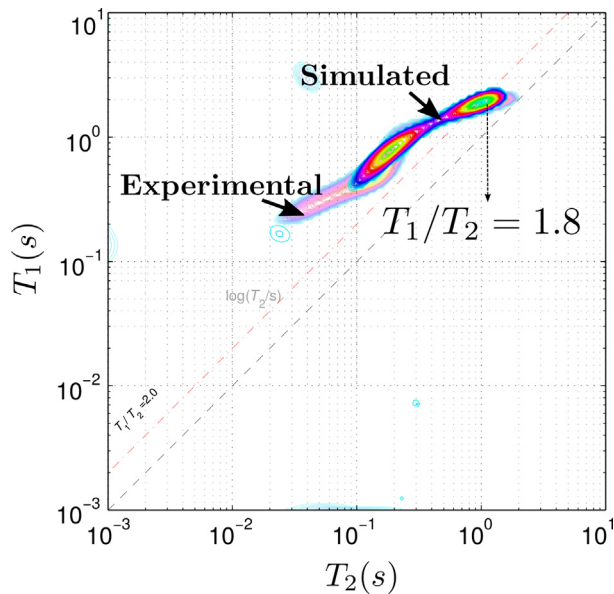


Fig. 9. The T_1 and T_2 bulk relaxation of water at room temperature have the same magnitude, and the correlation peak appears in the diagonal. As restrictions to the motion increase as the pore sizes decrease, the ratio between relaxation times T_1 and T_2 begin to move away from the diagonal. The restrictions imposed by the digital porous media reproduce in the simulations the same effect we see in the experiment. Using the T_1 to T_2 ratio for the larger pores we estimate the longitudinal relaxivity $\rho_1 = 6 \mu\text{m/s}$.

experimental relaxation time distribution was measured by CPMG, with 16,000 echoes and echo time of $150 \mu\text{s}$. We measured the magnetization evolution for different echo times, varying from 150 to $500 \mu\text{s}$ to analyze the influence of the internal magnetic field gradient. At 150 and $200 \mu\text{s}$, no significant T_2 changes were observed, but there were noticeable variations towards shorter T_2 for the longer echo times. So, we assume that the internal magnetic field gradient effects are negligible for the shorter echo times.

Fig. 10(a) shows the experimental and simulated relaxation time distributions. The agreement between the simulated and experimental results is justified because of better visualization of the pore morphology by micro-CT. The short T_2 peaks are defined by us as an eigenvalue of order 1 and 2, for also being present in the simulated relaxation time distribution and no pore with small size was identified in the Digital Porous Media. Fig. 10(b) shows the relaxation time distributions for different relaxivities varying from 10 to $50 \mu\text{m/s}$, or from the fast to the intermediate regime, respectively. The shorter T_2 appears only for $\rho_2 = 30 \mu\text{m/s}$, and increases in amplitude for larger relaxivities. The longer T_2 shifts a little for the different relaxivities and has the best fit with the experimental value for $\rho_2 = 50 \mu\text{m/s}$. The longer T_2 resulted in a pore size of $\sim 100 \mu\text{m}$, in agreement with the value measured directly in the Digital Porous Media.

5. Conclusions

We developed and validated a Random Walk method to simulate the fluid molecules dynamics inside a porous medium, in the macroscopic scale, and the corresponding effects on the NMR data. With the aid of micro-CT images and the above-described NMR data simulation, we presented an alternative method to estimate the surface relaxivity and, consequently, the pore size distribution. While the general methods employed to segment the 3D micro-CT images take into consideration only the geometric porous media features, our proposed method considers the molecular diffusion

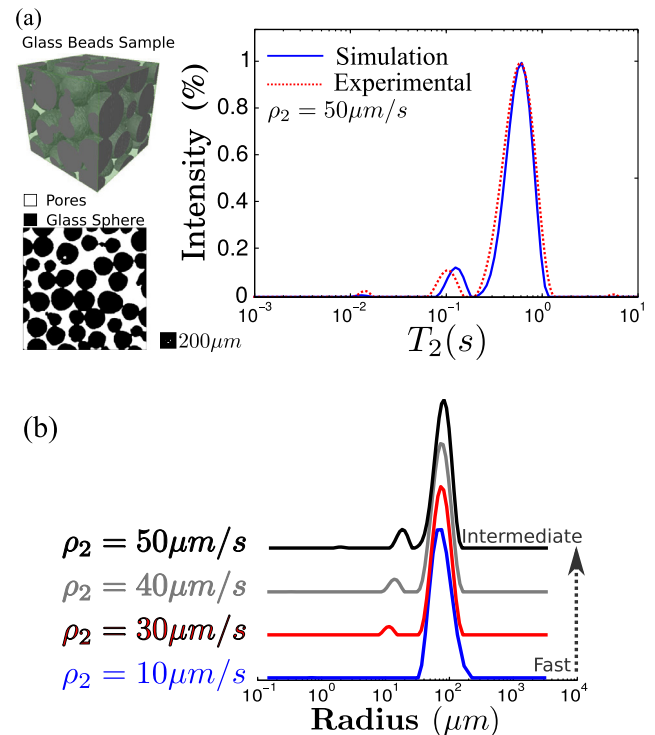


Fig. 10. (a) Three peaks appear in the simulation and experimental results for glass beads packing, with a good agreement. (b) Estimating the relaxivity we calculated the radii distribution and the $T_2 = 0.1 \text{ ms}$ have the respective radius equal $10 \mu\text{m}$. This pore size is not visualized by the micro-CT images. However, varying the surface magnetic relaxivity on the simulation from 10 to $50 \mu\text{m/s}$, we find that this peak is an eigenvalue contribution ($n = 1$).

dynamics within digital porous media, which gives appropriate correlation between NMR and pore size distribution.

These results indicate the potential of the simulations for exploring and better understanding the correlations between the NMR observables, molecular dynamics and the microstructure of the porous medium. As an example, the explicit correlation between collision number and pore size (hence surface relaxation times) expressed in Eqs. (21) and (22), along with the expression for $\bar{\lambda}$, allows a connection between the translational molecular dynamics ($\mathbf{M}$ found in the simulation) and T_2 distributions measured experimentally. Based on the porous media images one could, for example, recover geometry information by mapping collision numbers throughout the entire medium, and connecting them with T_2 distributions [31]. A similar approach that uses this kind of reasoning was reported in Ref. [32] where, using the Bergman model, the authors establish a relationship between the collision number mapping and variations in surface relaxivity.

The analysis of the pores radii using the general solution for the diffusion equation highlights the consideration we need to have when assuming the validity of the fast diffusion equation. The errors in estimating the radii when the relaxivity is known will also affect the estimation of the relaxivity, wettability or permeability from NMR techniques, which assumes the fast diffusion in the equations.

The synthetic materials are very important to validate the computational physics methods proposed, and we intend to produce other porous media with a wide range of pores visible by micro-CT, as well as different connections and surface relaxivities. We aim in further studies, with the simulations of the Exchange techniques and synthetic porous media, to correlate NMR with other microstructures parameters and, consequently, the permeability properties of porous media [5,6].

Acknowledgments

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References

- [1] P. McDonald, J.-P. Korb, J. Mitchell, L. Monteilhet, Surface relaxation and chemical exchange in hydrating cement pastes: a two-dimensional NMR relaxation study, *Phys. Rev. E* 72 (2005) 011409, <https://doi.org/10.1103/PhysRevE.72.011409>.
- [2] Y.-Q. Song, Magnetic resonance of porous media (MRPM): a perspective, *J. Magn. Reson.* 229 (2013) 12–24, <https://doi.org/10.1016/j.jmr.2012.11.010>.
- [3] L. Brizi, M. Barbieri, F. Baruffaldi, V. Bortolotti, C. Fersini, H. Liu, M. Nogueira d'Eurydice, S. Obruchkov, F. Zong, P. Galvosas, Bone volume-to-total volume ratio measured in trabecular bone by single-sided NMR devices, *Magn. Reson. Med.* (2017), <https://doi.org/10.1002/mrm.26697>.
- [4] K. Washburn, P. Callaghan, Tracking pore to pore exchange using relaxation exchange spectroscopy, *Phys. Rev. Lett.* 97 (2006) 175502, <https://doi.org/10.1103/PhysRevLett.97.175502>.
- [5] R. Dortch, R. Horsch, M. Does, Development, simulation, and validation of NMR relaxation-based exchange measurements, *J. Chem. Phys.* 131 (2009) 164502, <https://doi.org/10.1063/1.3245866>.
- [6] M. Van Landeghem, A. Haber, J.B. D'espinoze De Lacaille, B. Blümich, Analysis of multisite 2D relaxation exchange NMR, *Concepts in Magnetic Resonance Part A*, 36 (2010) 153–169, <http://doi.org/10.1002/cmr.a.20157>.
- [7] Y.-Q. Song, L. Venkataramanan, M. Hürlimann, M. Flaum, P. Frulla, C. Straley, T1–T2 correlation spectra obtained using a fast two-dimensional Laplace inversion, *J. Magn. Reson.* 154 (2002) 261–268, <https://doi.org/10.1006/jmre.2001.2474>.
- [8] M. Hürlimann, L. Venkataramanan, Quantitative measurement of two-dimensional distribution functions of diffusion and relaxation in grossly inhomogeneous fields, *J. Magn. Reson.* 157 (2002) 31–42, <https://doi.org/10.1006/jmre.2002.2567>.
- [9] J.L. Paulsen, Y.-Q. Song, Two-dimensional diffusion time correlation experiment using a single direction gradient, *J. Magn. Reson.* 244 (2014) 6–11, <https://doi.org/10.1016/j.jmr.2014.04.007>.
- [10] K.R. Brownstein, C. Tarr, Importance of classical diffusion in NMR studies of water in biological cells, *Phys. Rev. A* 19 (1979) 2446, <https://doi.org/10.1016/j.jmr.2014.04.007>.
- [11] P.P. Mitra, P.N. Sen, L.M. Schwartz, P. Le Doussal, Diffusion propagator as a probe of the structure of porous media, *Phys. Rev. Lett.* 68 (1992) 3555, <https://doi.org/10.1103/PhysRevLett.68.3555>.
- [12] L.L. Latour, P.P. Mitra, R.L. Kleinberg, C.H. Sotak, Time-dependent diffusion coefficient of fluids in porous media as a probe of surface-to-volume ratio, *J. Magn. Reson., Ser. A* 101 (1993) 342–346, <https://doi.org/10.1002/cmr.a.20017>.
- [13] K.S. Mendelson, Percolation model of nuclear magnetic relaxation in porous media, *Phys. Rev. B* 41 (1990) 562, <https://doi.org/10.1103/PhysRevB.41.562>.
- [14] D.J. Bergman, K.-J. Dunn, L.M. Schwartz, P.P. Mitra, Self-diffusion in a periodic porous medium: a comparison of different approaches, *Phys. Rev. E* 51 (1995) 3393, <https://doi.org/10.1103/PhysRevE.51.3393>.
- [15] C.H. Arns, A comparison of pore size distributions derived by NMR and X-ray-CT techniques, *Phys. A: Statist. Mech. Appl.* 339 (2004) 159–165, <https://doi.org/10.1016/j.physa.2004.03.033>.
- [16] C. Arns, Y. Melean, L. Burcaw, P. Callaghan, K. Washburn, Comparison of experimental NMR measurements with simulated responses on digitized images of mono-mineralic rocks using Xray-CT, in: *Improved Core Analysis for Unconventional Fields*, number SCA2009-44 in *The 23rd International Symposium of the Society of Core Analysts*, 2009, pp. 1–6.
- [17] I. Shikhov, M.N. d'Eurydice, J.-Y. Arns, C.H. Arns, An experimental and numerical study of relative permeability estimates using spatially resolved T₁-z NMR, *Transp. Porous Media* (2017) 1–26, <https://doi.org/10.1007/s11242-017-0855-7>.
- [18] F. Reif, *Fundamentals of Statistical and Thermal Physics*, Waveland Press, 2009.
- [19] H.Y. Carr, E.M. Purcell, Effects of diffusion on free precession in nuclear magnetic resonance experiments, *Phys. Rev.* 94 (1954) 630, <https://doi.org/10.1103/PhysRev.94.630>.
- [20] M. Hürlimann, L. Latour, C. Sotak, Diffusion measurement in sandstone core – NMR determination of surface-to-volume ratio and surface relaxivity, *Magn. Reson. Imaging* 12 (1994) 325–327, [https://doi.org/10.1016/0730-725X\(94\)91548-2](https://doi.org/10.1016/0730-725X(94)91548-2).
- [21] H.C. Torrey, Bloch equations with diffusion terms, *Phys. Rev.* 104 (1956) 563, <https://doi.org/10.1103/PhysRev.104.563>.
- [22] N. Bloembergen, E.M. Purcell, R.V. Pound, Relaxation effects in nuclear magnetic resonance absorption, *Phys. Rev.* 73 (1948) 679, <https://doi.org/10.1103/PhysRev.73.679>.
- [23] R. Kleinberg, W. Kenyon, P. Mitra, Mechanism of NMR relaxation of fluids in rock, *J. Magn. Reson., Ser. A* 108 (1994) 206–214, <https://doi.org/10.1006/jmra.1994.1112>.
- [24] J.-P. Korb, M. Whaley-Hodges, R. Bryant, Translational diffusion of liquids at surfaces of microporous materials: theoretical analysis of field-cycling magnetic relaxation measurements, *Phys. Rev. E* 56 (1997) 1934, <https://doi.org/10.1103/PhysRevE.56.1934>.
- [25] P.-Z. Wong, T. Lucatorto, M. De Graef, *Methods of the Physics of Porous Media*, Academic Press, 1999.
- [26] A. Buades, B. Coll, J.-M. Morel, A non-local algorithm for image denoising, in: *Computer Vision and Pattern Recognition, 2005. CVPR 2005. IEEE Computer Society Conference on*, IEEE, 2005, pp. 60–65, <http://doi.org/10.1109/CVPR.2005.38>.
- [27] C.P. Slichter, *Principles of magnetic resonance*, Springer Series in Solid-State Sciences, 1 (1990) 497. ISBN: 9783642080692.
- [28] F. Bloch, Nuclear induction, *Phys. Rev.* 70 (1946) 460, <https://doi.org/10.1103/PhysRev.70.460>.
- [29] R.W. Brown, Y.-C.N. Cheng, E.M. Haacke, M.R. Thompson, R. Venkatesan, *Magnetic Resonance Imaging: Physical Principles and Sequence Design*, John Wiley & Sons, 2014.
- [30] M.N. d'Eurydice, E.T. Montrazi, C.A. Fortulan, T.J. Bonagamba, T2-filtered T2–T2 exchange NMR, *J. Chem. Phys.* 144 (2016) 204201, <https://doi.org/10.1063/1.4951712>.
- [31] E.J. Fordham, J. Mitchell, Localization in a single pore, *Microporous Mesoporous Mater.* (2017), <https://doi.org/10.1016/j.micromeso.2017.05.029>.
- [32] F. Benavides, R. Leiderman, A. Souza, G. Carneiro, R. Bagueira, Estimating the surface relaxivity as a function of pore size from NMR T2 distributions and micro-tomographic images, *Comput. Geosci.* (2017), <https://doi.org/10.1016/j.cageo.2017.06.016>.